

# AUTHBC — Running Results Summary

Co-optimizing encoding  $\times$  authentication placement  $\times$  signature scheme  $\times$  batching  
for blockchain-grade UAV telemetry over 802.11

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(living document — regenerated as each step completes)

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## Abstract

This document tracks every measured result in the AUTHBC project and is updated at each step. All numbers trace to a frozen CSV in **results/** and are guarded by the reproduction gate (**make verify-frozen**). The headline is established: the co-optimized configuration cuts on-air authentication bytes by **96.4 %** versus the inline-CBOR baseline while holding verifiability  $V \geq 0.95$  at loss  $p = 0.05$  — criterion met. The hardware campaign has confirmed the one predicted anomaly (the signature-scheme ordering flips on ARM) and has shown the energy model’s assumed CPU power to be  $4.1\times$  too high.

## 1 The problem

Telemetry records are tiny (45–190 B) but a signature is 64–96 B, so authentication overhead rivals the payload. Four knobs — record *encoding*, authentication *placement* (A inline, B self-batch, C relay-aggregate, D block), signature *scheme*, and *batch* size  $b$  — are coupled and must be co-optimized. The falsifiable criterion: the co-optimized configuration must cut on-air authentication bytes by  $\geq 40\%$  versus inline-CBOR at  $V \geq 0.95$ ,  $p = 0.05$ .

## 2 Software layer (frozen, reproducible)

### 2.1 E1 — overhead dominance (T1)

$\varphi = g/(s + g)$  is the fraction of on-air bytes that is pure signature ( $g = 64\text{ B}$ ).

| encoding     | mean record $s$ | $\varphi$     |
|--------------|-----------------|---------------|
| JSON         | 191.1 B         | 25.1 %        |
| CBOR         | 66.25 B         | 49.1 %        |
| MessagePack  | 65.16 B         | 49.6 %        |
| <b>delta</b> | <b>45.00 B</b>  | <b>58.7 %</b> |

**Reading.** The better the compression, the *worse* the relative overhead: delta is the best encoder and leaves 59 % of the frame as signature. Compression alone cannot solve this — the motivation for the other three knobs.

### 2.2 E2 — batching cure (T2)

Folding  $b$  records under one signature gives  $s + (g_a + H_f)/b$  per record; packing to the MTU bottoms out at  $A = M/(M - H_f - g_a) = 1.0745$  at  $M = 1500$ . Measured: delta reaches  $b_{\max} = 31$  and attains  $A = 1.0745$  exactly; other encodings within 5 % (integer- $b$  slack). Inline placement *never* amortizes its per-record signature — only the header shrinks.

### 2.3 E3 — loss frontier (T3)

Self-batch frames verify independently, so  $V_B = 1 - p$ , *flat in b*. A block signature spanning  $n$  frames needs all fragments:  $V_D = (1 - p)^n$ . Measured at  $p = 0.10$ ,  $b = 40$ :  $V_B = 0.90$  versus  $V_D = 0.81 = 0.9^2$ . This is the wall that stops over-aggregation.

### 2.4 E4 — scheme crossover (T4)

BLS aggregates but its pairing verify is  $10.7\times$  costlier than Ed25519. Which wins depends on the powers only through  $\kappa = P_r/P_c$ , so the break-even  $\kappa^*$  is found without measuring watts. **Result: Ed25519 wins all 80 grid points**,  $\min \kappa^* = 3.21$ . At the accepted 96 B, BLS carries *more* bytes than Ed25519's 64 B on own traffic ( $\kappa^* = \infty$ ); its aggregation pays off only for cross-signer relay traffic.

### 2.5 E5 — the headline (T5) **PASS**

| config            | enc   | scheme  | placement | auth B/rec  | $V$                |
|-------------------|-------|---------|-----------|-------------|--------------------|
| <b>optimized</b>  | delta | ECDSA   | B, $b=28$ | <b>3.71</b> | 0.95               |
| A+CBOR (baseline) | cbor  | Ed25519 | A inline  | 104.0       | 0.95               |
| A+JSON (naive)    | json  | Ed25519 | A inline  | 104.0       | 0.95               |
| D-over-agg        | cbor  | Ed25519 | D, $b=40$ | 3.60        | <b>0.9025 FAIL</b> |

$$\text{cut} = \frac{104 - 3.71}{104} = \mathbf{96.4\%} \geq 40\% \Rightarrow \mathbf{PASS}$$

Hand-checked:  $45.0 + 104/28 = 48.71$  B/record. **The last row is the co-design argument in miniature:** over-aggregation is byte-competitive (3.60 B) yet *fails*  $V \geq 0.95$  because  $0.9025 = (1 - p)^2$ . The byte-optimal answer is not the correct answer.

### 2.6 NS-3 channel validation

| $N$ | unicast gap  | broadcast gap |
|-----|--------------|---------------|
| 5   | 5.1 %        | 4.4 %         |
| 20  | 5.2 %        | 37 %          |
| 50  | <b>1.8 %</b> | 1735 %        |

Unicast validates the Bianchi DCF model to  $\pm 1.8\text{--}5.3\%$ . The broadcast column is discussed in §4 — part real capture effect, part an artefact of node placement.

## 3 Hardware layer (Raspberry Pi 4B, measured)

### 3.1 Cross-platform determinism

All **12/12** encoded-size metrics are byte-identical between x86 and ARM, as deterministic integer-only encoders require.

### 3.2 Timings and cycle normalisation

| op             | x86          | RPi4   | wall ratio    | <b>cycles ARM/x86</b>          |
|----------------|--------------|--------|---------------|--------------------------------|
| ed25519 sign   | 29.1 $\mu$ s | 88.1   | 3.03 $\times$ | <b>1.16<math>\times</math></b> |
| ed25519 verify | 95.0         | 259.5  | 2.73 $\times$ | <b>1.05<math>\times</math></b> |
| ecdsa sign     | 26.2         | 135.5  | 5.17 $\times$ | 1.98 $\times$                  |
| ecdsa verify   | 78.5         | 326.9  | 4.16 $\times$ | 1.60 $\times$                  |
| bls verify     | 1016.5       | 8611.8 | 8.47 $\times$ | 3.25 $\times$                  |

Wall-clock ratio conflates clock speed with per-cycle efficiency. The clock floor is  $4.7/1.8 = 2.61\times$ . Normalising shows Ed25519 needs only  $1.05\times$  the cycles on ARM (equally optimised on both ISAs) while BLS needs  $3.25\times$  (heavily x86-tuned). **Consequence: the docs/04 §1 “5–15 $\times$ ” anchor is unreachable** for an equally-optimised primitive and must be restated per-cycle.

### 3.3 F6 — the scheme ordering flips on ARM

| platform    | ECDSA verify | Ed25519 verify | winner         |
|-------------|--------------|----------------|----------------|
| x86         | 78.5 $\mu$ s | 95.0 $\mu$ s   | ECDSA          |
| <b>RPi4</b> | <b>326.9</b> | <b>259.5</b>   | <b>Ed25519</b> |

Three candidate causes were separated:

- **Library — ruled out.** Byte-identical versions on both (cryptography 45.0.7).
- **Our wrapper — real but not the cause.** Our ECDSA path does ASN.1 DER conversion in Python every op (Ed25519 does not): 1.05  $\mu$ s on x86, **12.58  $\mu$ s on ARM** ( $\sim 12\times$  worse). Flipping the result would have required  $64\times$ . Core-only on ARM: ECDSA 312.4 versus Ed25519 263.3 — **Ed25519 still wins**.
- **Hardware — the operative cause**, but see §4: the *specific* microarchitectural mechanism is **not** established.

**Consequence:** on the real deployment platform the optimum selects **Ed25519** (the D2 architectural default). Both are 64B, so the 96.4% headline is **unchanged**.

### 3.4 Measured per-op energy (RPi4, 60 s windows, 5 reps)

| op             | energy/op ( $\mu$ J) | 95 % CI         | $\Delta P$ (W) |
|----------------|----------------------|-----------------|----------------|
| ed25519 sign   | <b>70.62</b>         | 70.11–71.24     | 0.771          |
| ed25519 verify | <b>186.84</b>        | 184.37–188.38   | 0.712          |
| ecdsa sign     | 87.29                | 86.73–88.69     | 0.630          |
| ecdsa verify   | 203.51               | 201.77–206.80   | 0.617          |
| bls sign       | 1464.61              | 1442.80–1489.06 | 0.514          |
| bls verify     | 5121.91              | 5068.33–5205.19 | 0.520          |
| delta encode   | 36.19                | 35.89–36.60     | 0.726          |
| cbor encode    | 40.11                | 39.65–40.58     | 0.753          |

**Ed25519 is cheaper than ECDSA in energy as well as time on ARM** (70.6 vs 87.3  $\mu$ J sign; 186.8 vs 203.5 verify), consistent with the timing flip. Independent cross-check against  $\Delta P \times t_{op}$  (micro suite) agrees within 1–4.4% for seven of eight ops; **bls:verify is 14.4% off** and is flagged for investigation under Law 7 (sustained 60 s loop versus the micro suite’s short bursts is the leading hypothesis). Measured  $p_{cpu}$  median **0.712 W** (range 0.514–0.771).

### 3.5 Reproducibility and energy

Two boards, independent runs: worst-case divergence **0.5%** across all 16 timed ops. The energy chain validated end-to-end (12 windows  $\rightarrow$  12 segments) with two independent cross-checks (loop throughput versus micro suite within 5–6%; single-core  $\Delta P$  0.728 W versus 4-core stress 0.690 W/core). **Measured  $p_{cpu} \approx 0.73$  W against the nominal 3.0 W used in E5** —  $4.1\times$  **too high**. The auth-byte headline is power-free and unaffected.

## 4 Open items and inconsistencies

| item                        | status                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F6 mechanism                | <b>Claim retracted.</b> Masking ADX/BMI2 on x86 via <code>OPENSSL_ia32cap</code> did <i>not</i> slow ECDSA ( $61.1 \rightarrow 56.8 \mu\text{s}$ ; ECDSA still won), so the ADX explanation is <b>unsupported</b> . What remains established: the ordering is ISA-dependent and Ed25519 is markedly more portable ( $1.05\times$ versus $1.60\times$ cycles). The specific mechanism is an open question. |
| NS-3 broadcast              | Nodes sit on a <i>line</i> at 1 cm spacing under a log-distance model ( $n = 3$ , $d_0 = 1 \text{ m}$ ), giving a <b>50.7 dB</b> near-far power spread. Capture needs only a few dB, so the geometry manufactures the very effect blamed for the gap. Needs an equal-power channel to test the model on its own assumptions.                                                                              |
| $\kappa$ plausible band     | E4 assumes $\kappa = P_r/P_c \lesssim 0.5$ from “radio below CPU power”. With $p_{cpu}$ measured at 0.73 W (not 3.0 W), $\kappa$ could approach or exceed 1. The E4 verdict is unchanged ( $\min \kappa^* = 3.21$ ) but the stated band must be re-derived from measured powers.                                                                                                                          |
| $p_{radio}$<br>E4/E5 re-run | <b>pending</b> — needs the 2-node link run; both boards now healthy.<br><b>pending</b> — deliberately deferred until both powers are measured, so the energy column stays internally consistent rather than half-measured.                                                                                                                                                                                |
| 5–15 $\times$ anchor        | Needs restating per-cycle (decision required).                                                                                                                                                                                                                                                                                                                                                            |

## Provenance

Every number above traces to a frozen CSV under `results/`; derived artefacts are regenerated byte-identically by `make verify-frozen`, which fails loudly on any drift. Full audit: `docs/audits/p7.md`; decisions and their limitations: `docs/DECISIONS.md`.